

GuardianGarment: Step-by-Step POC Guide

Phase 1: Hardware Procurement

Components: ESP32-C3 microcontroller, 3.7V LiPo battery, TP4056 charging module, COB LED array, black vinyl/cardstock for the stencil, and a thin, waterproof TPU pouch.

Phase 2: The BLE Handshake (Firmware)

Step 1: Whitelist Configuration

Program the ESP32-C3 to operate in "Scanner" mode. Hard-code the MAC address of the caregiver's mobile phone into the "Whitelist" array [cite: 3].

Step 2: Signal Monitoring

Implement a continuous loop that polls for the presence of the Whitelisted MAC address [cite: 3].

Step 3: Debounce Logic

Configure a timer variable to wait for 60–120 seconds if the signal is lost. If the signal returns within this window, reset the timer to prevent false alarms [cite: 3].

Phase 3: The Pulsing Reveal Mechanism

Step 1: Stencil Fabrication

Create a two-row stencil. Row 1: "PERSON WITH MEMORY LOSS – PLEASE HELP". Row 2: "CALL: [MOBILE NUMBER]". Use high-contrast cutouts on a dark, durable fabric [cite: 3].

Step 2: LED Array Integration

Align the LED array directly behind the stencil. Ensure a white diffuser layer is placed between the LEDs and the stencil to prevent hot spots [cite: 3].

Step 3: Pulsing Sequence

Update firmware to initiate a Pulse Width Modulation (PWM) or simple blink loop (0.5s On/0.5s Off) when the Debounce timer triggers an event [cite: 3].

Phase 4: Modular Assembly

Mount the ESP32, battery, and LED assembly into the TPU pouch. Attach this pouch to the T-shirt using a Velcro or snap-button base for easy removal and cleaning [cite: 3].

Phase 5: Field Verification

Verification 1: The Handshake

Ensure the LEDs remain off while the caregiver's phone is within range.

Verification 2: The Separation Trigger

Move the caregiver's phone out of range. Confirm the pulse starts after the 60-second debounce period [cite: 3].

Verification 3: The Find Function

While the stencil is pulsing, open a BLE scanner app on your phone. Identify the device name and confirm signal strength changes as you move closer to the T-shirt [cite: 3].